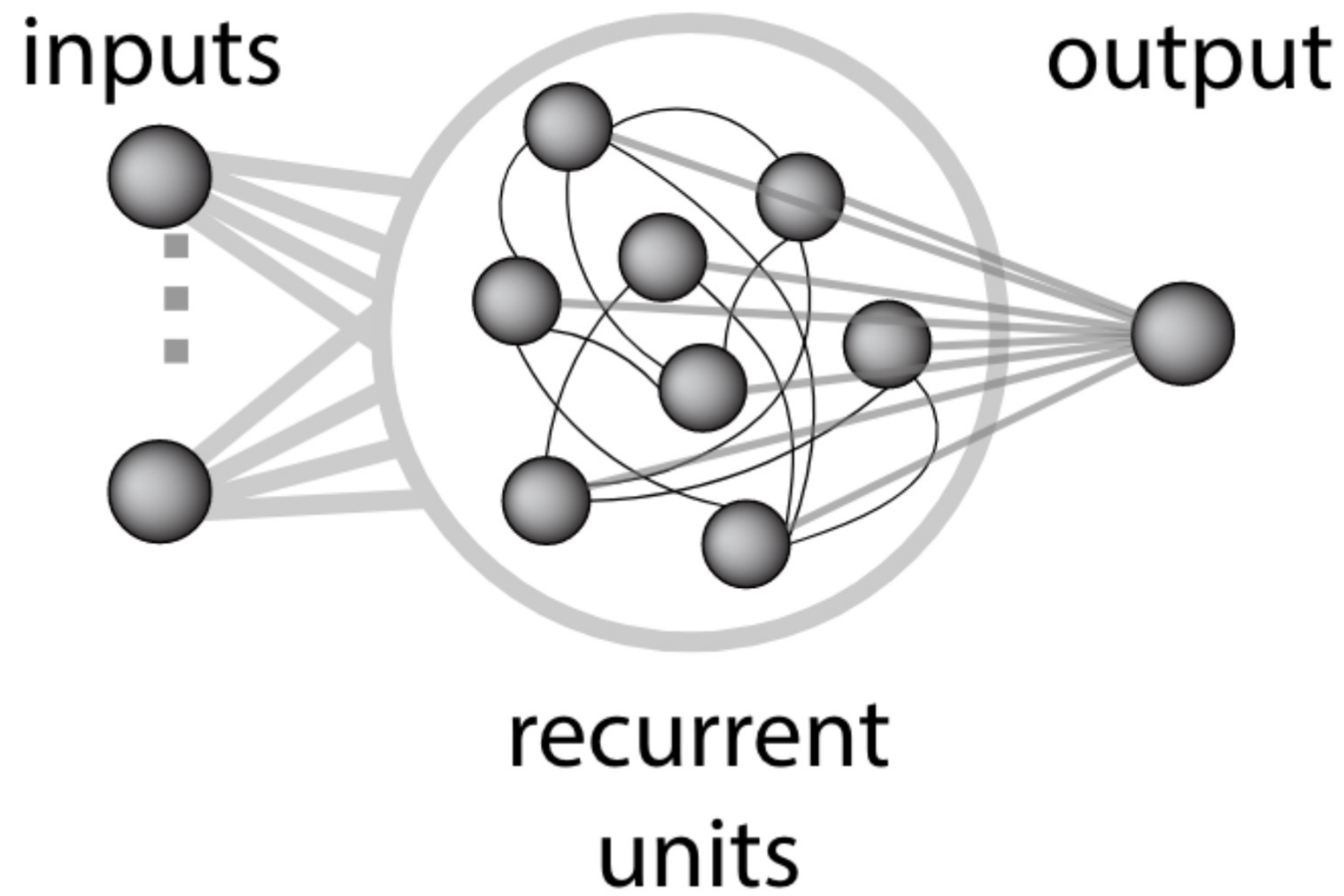
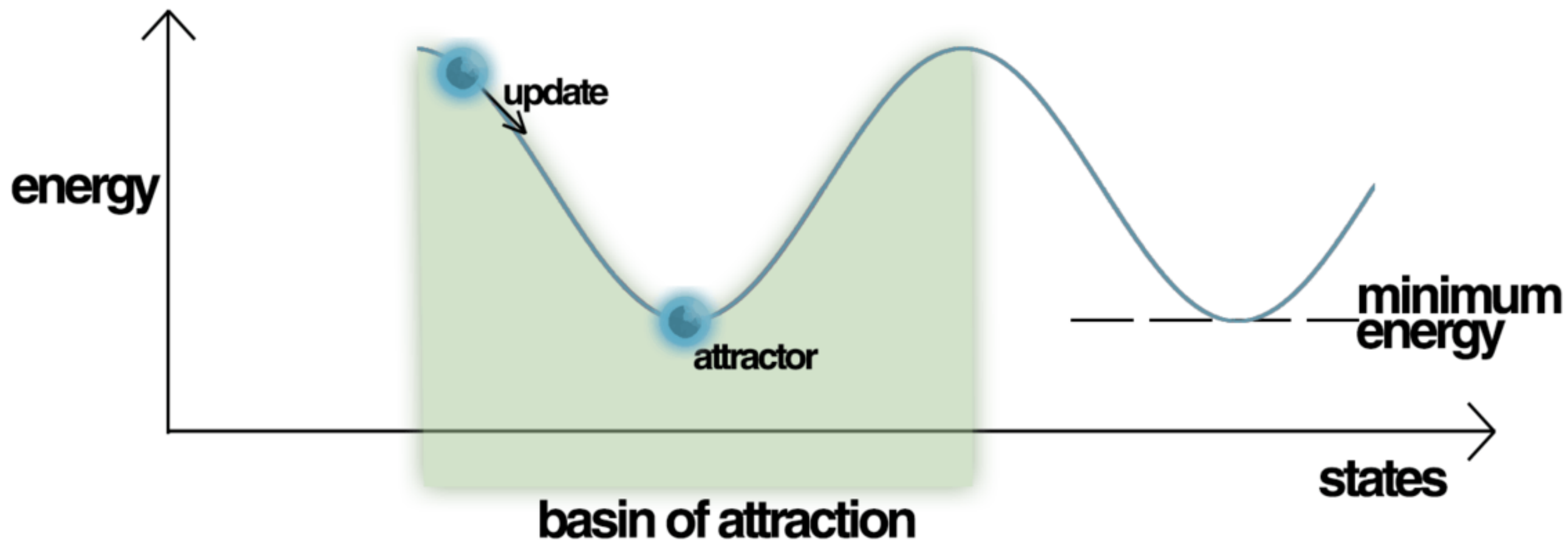


Attractor dynamics

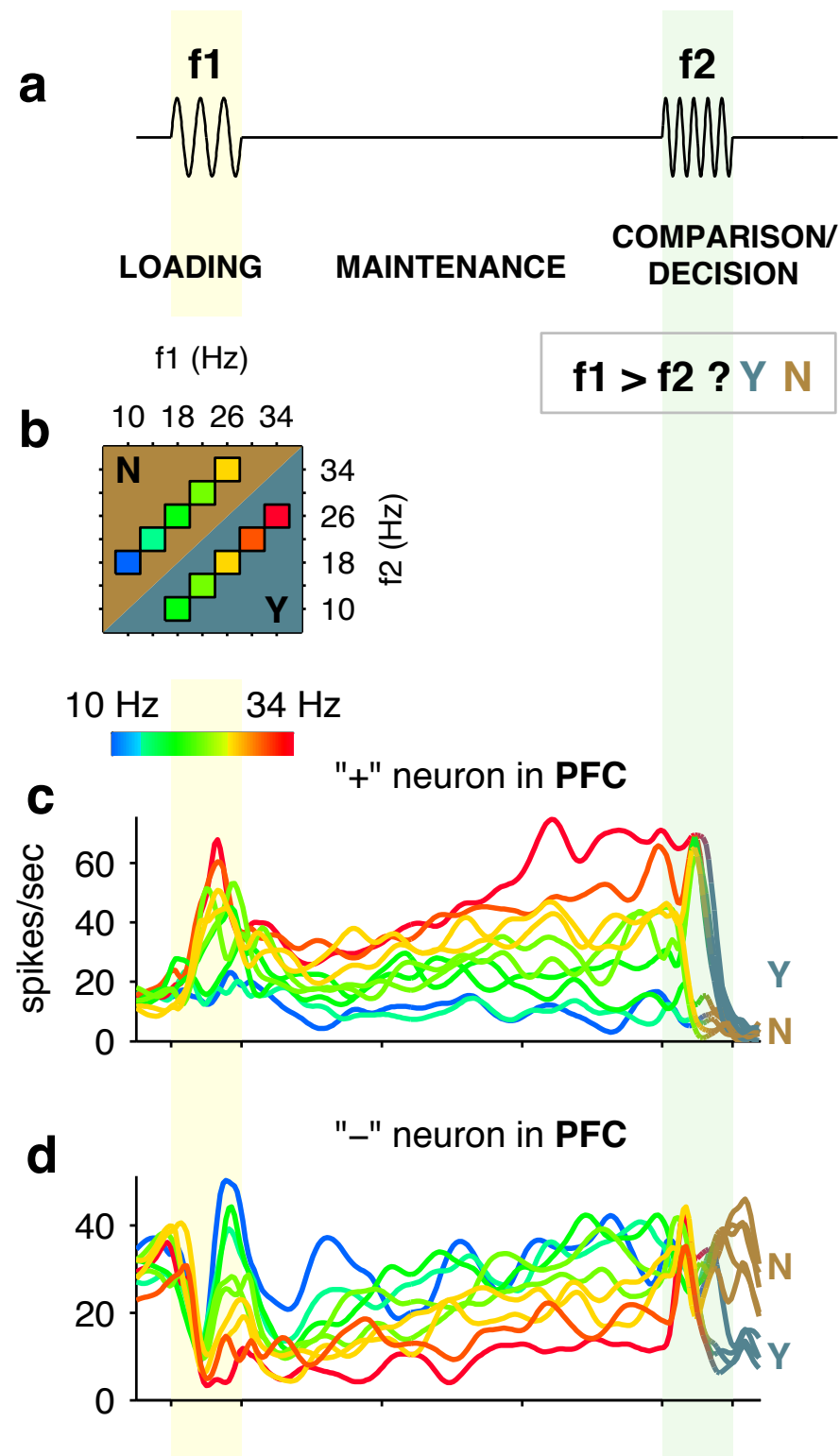
RNN



$$\frac{d\mathbf{v}}{dt} = -\nabla E$$

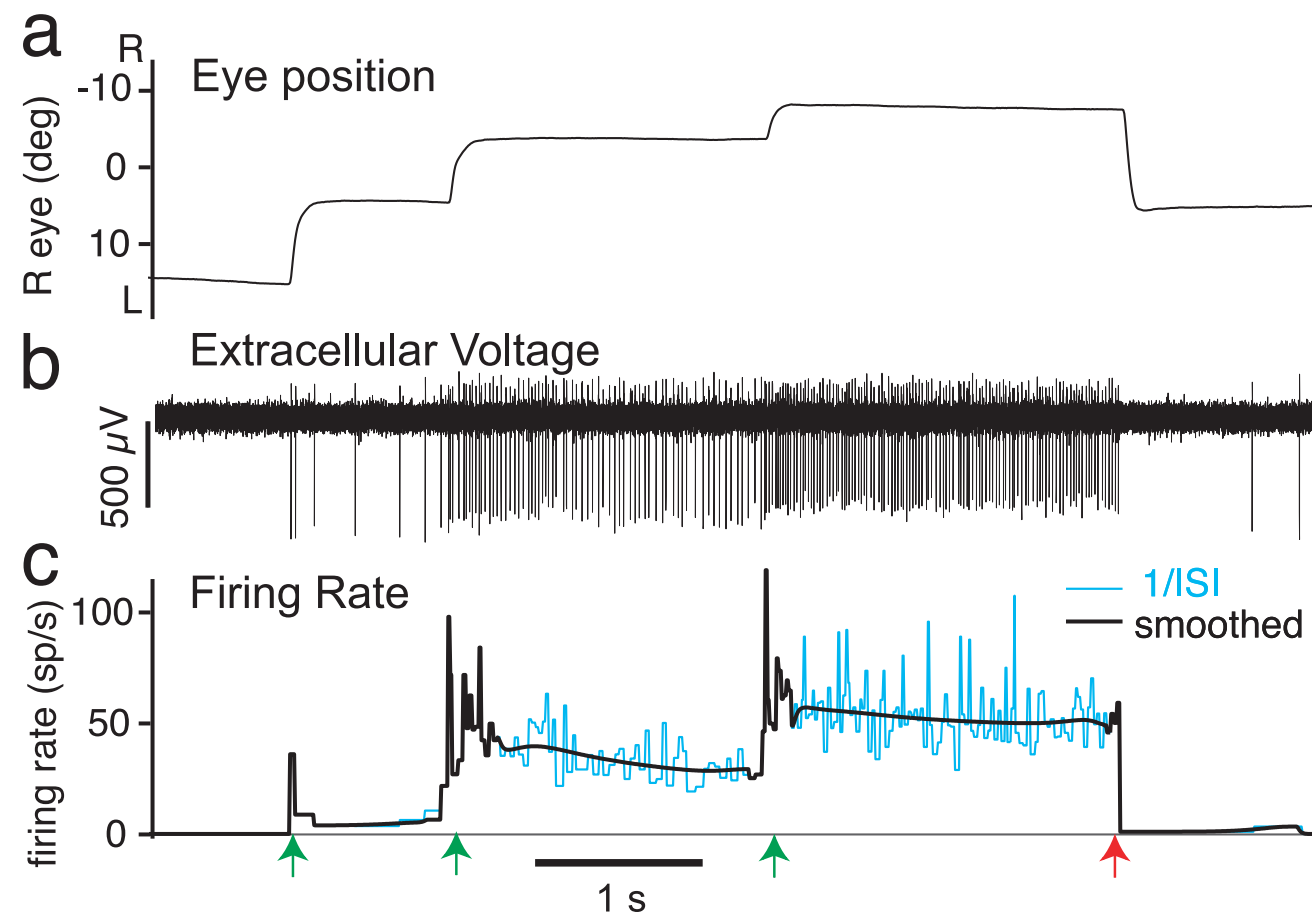


A parametric working memory task

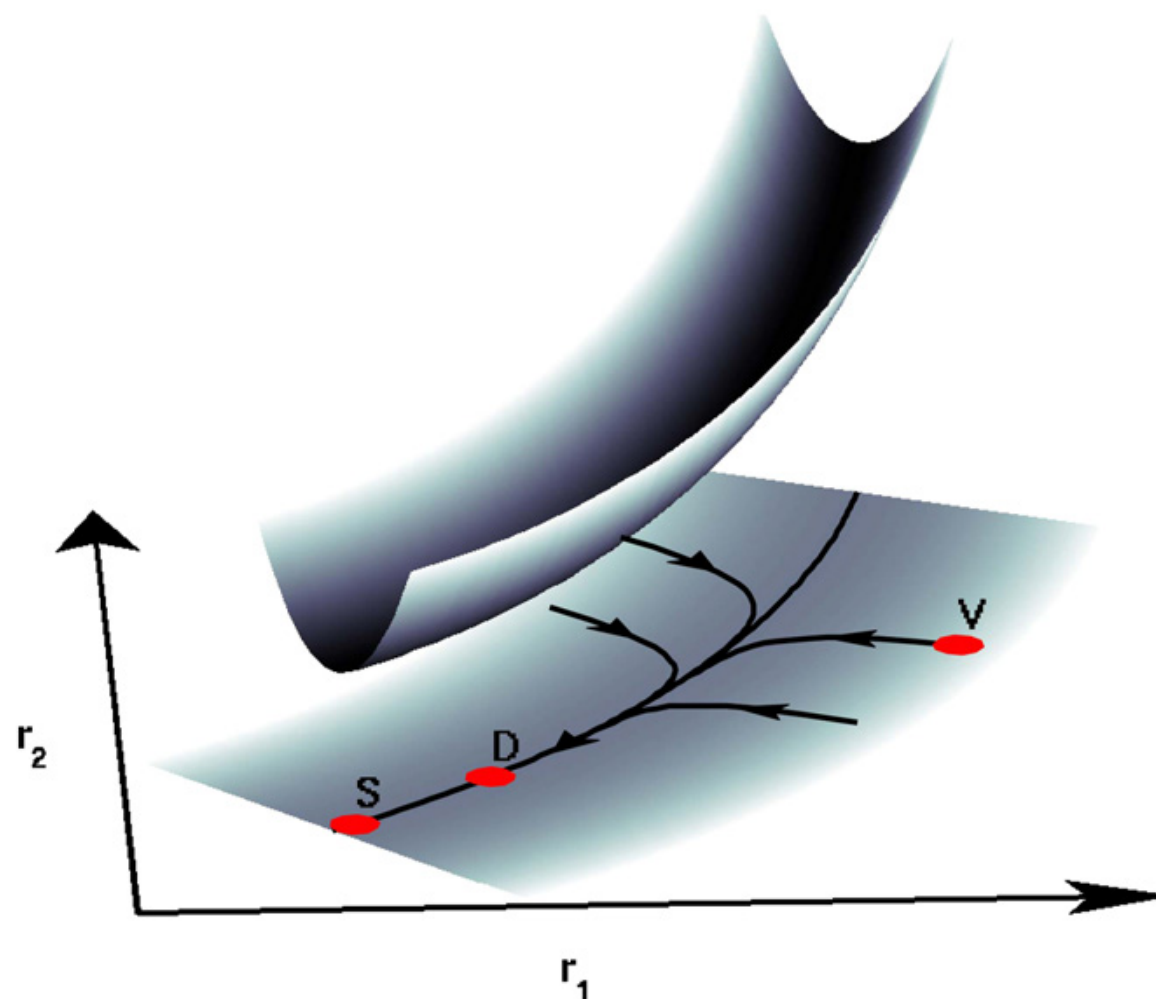


Romo et al 1999

Persistent activity in neurons encoding eye position



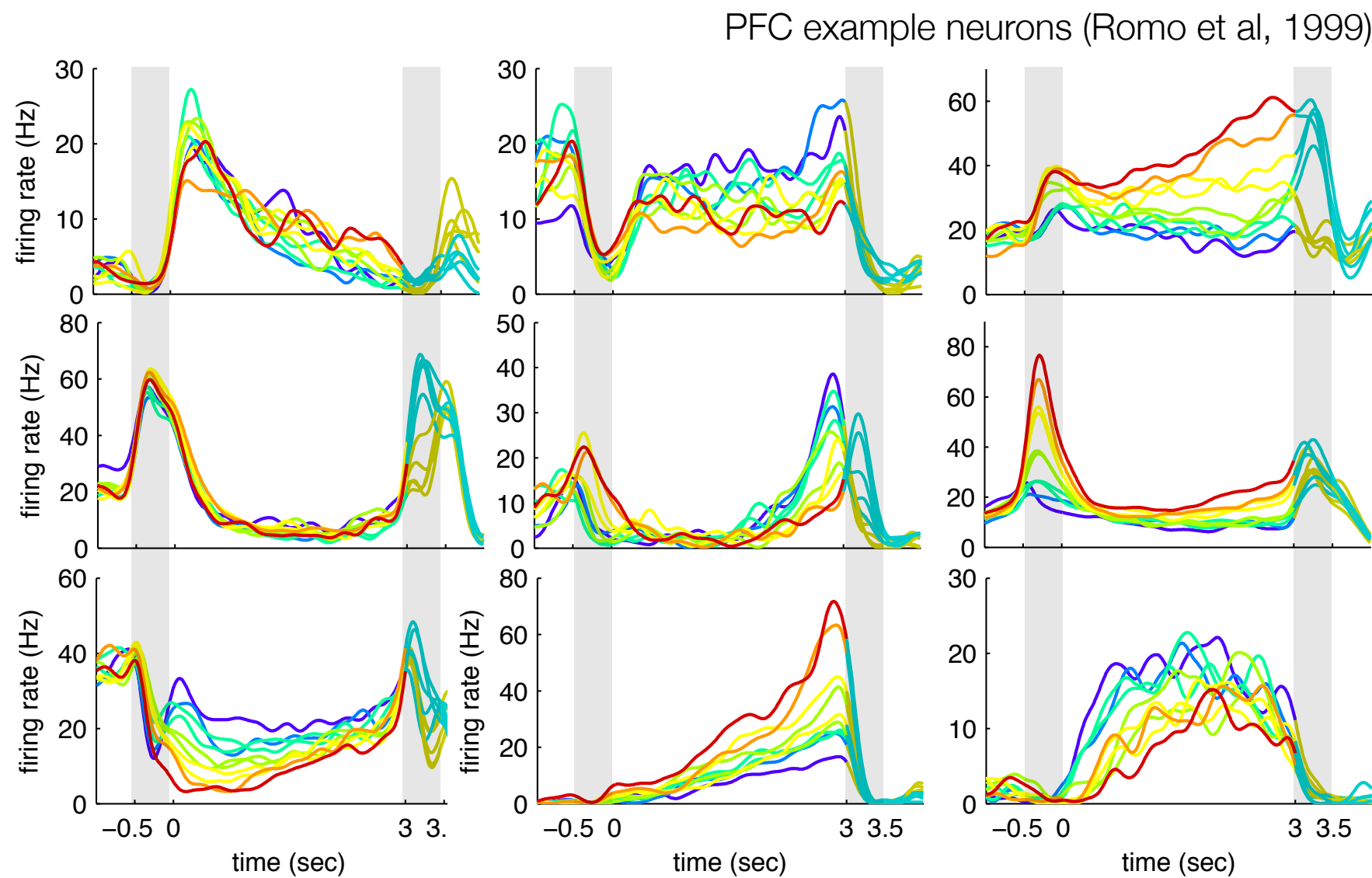
One dimensional dynamics



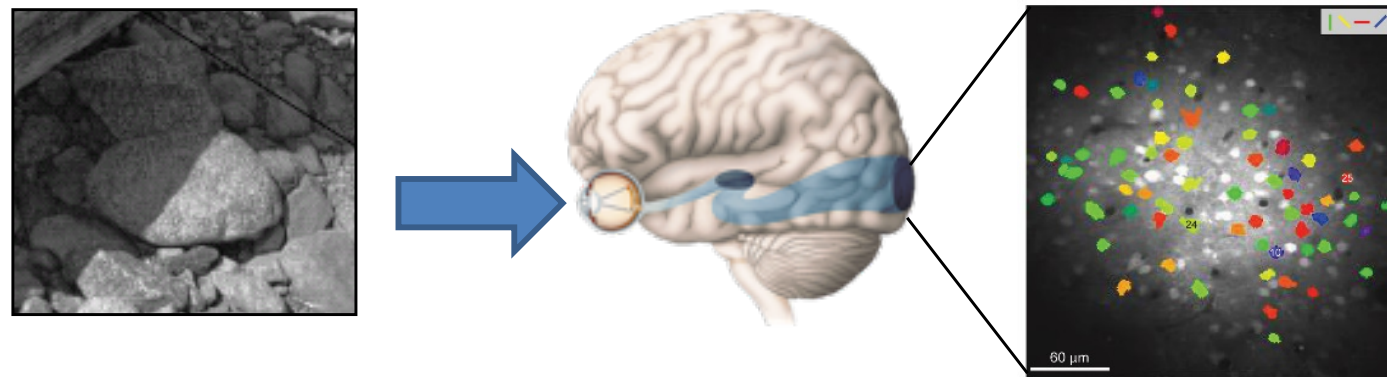
Mixed Selectivity

PFC, working memory task

PSTHs of nine (out of ~1000) cells

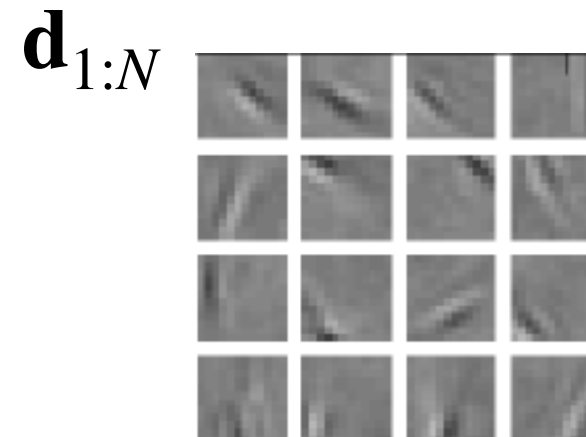


Working memory and over-representation



$$\mathbf{s} \approx \sum_{i=1}^N r_i(t) \mathbf{d}_i$$

$$\frac{ds}{dt} = 0$$



Fixed points and stability

Global stability: starting from **any** initial condition

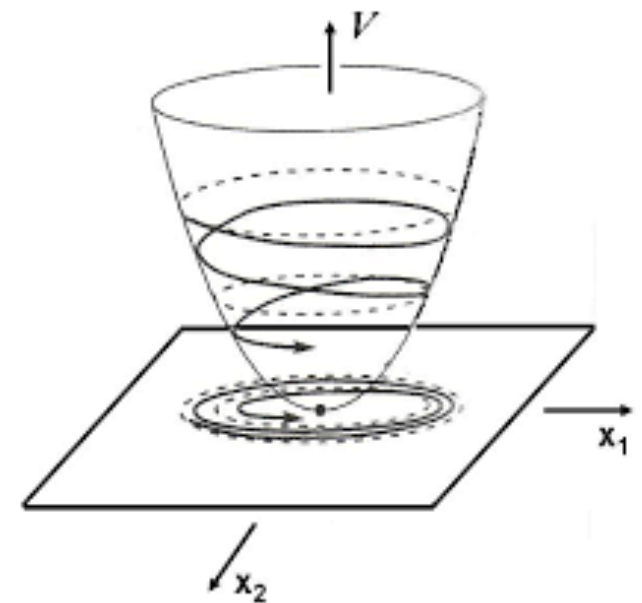
$$\frac{d\mathbf{u}}{dt} = \mathbf{f}(\mathbf{u})$$

Lyapunov function $V(u)$

$V(u) \geq V_0$, has a lower bound

$$\frac{d}{dt}V(u(t)) \leq 0$$

$$\frac{d}{dt}V(u(t)) = 0 \Rightarrow \nabla V(u) = 0$$



Theorem: If there exists a Lyapunov function, the system is (globally) stable where the trajectory will converge to one of the extrema of $V(u)$.

An example

$$\frac{dx}{dt} = -x + 4y$$

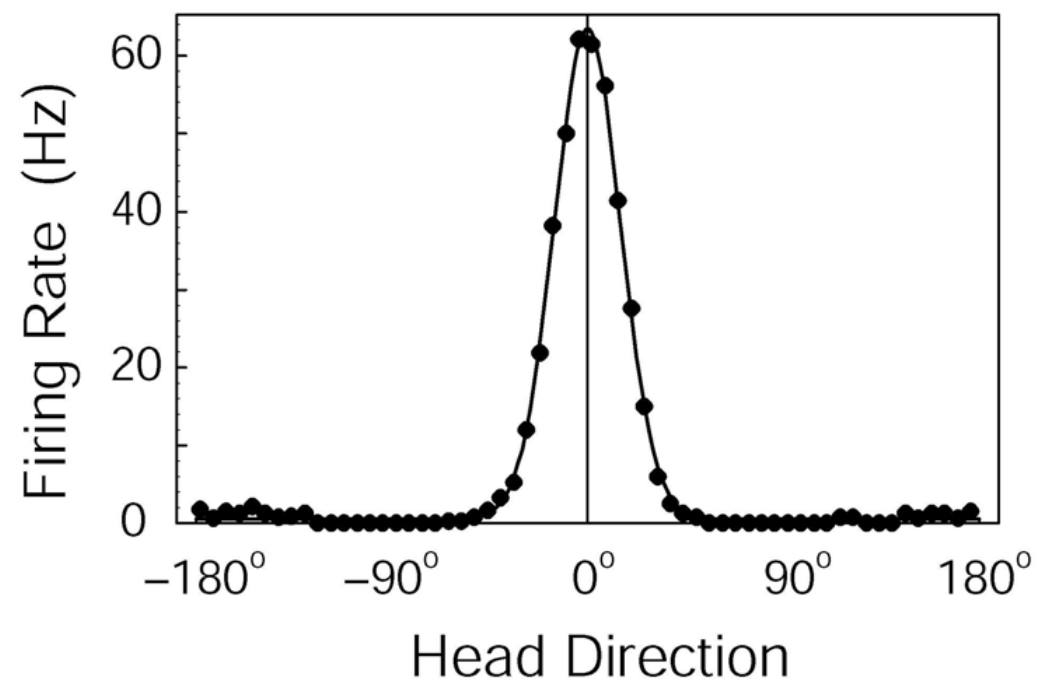
$$\frac{dy}{dt} = -x - y^3$$

Consider $V(x, y) = x^2 + \alpha y^2$

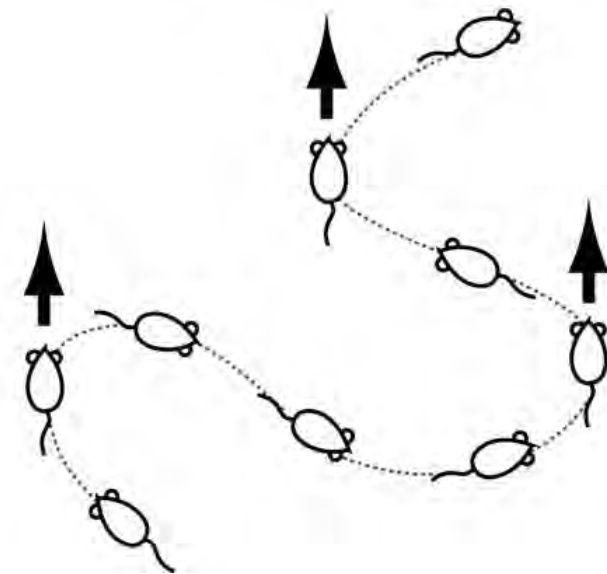
$$\begin{aligned}\dot{V} &= 2x\dot{x} + 2\alpha y\dot{y} = 2x(-x + 4y) + 2\alpha y(-x - y^3) \\ &= -2x^2 - 2\alpha y^4 + (8 - 2\alpha)xy\end{aligned}$$

Head direction cell

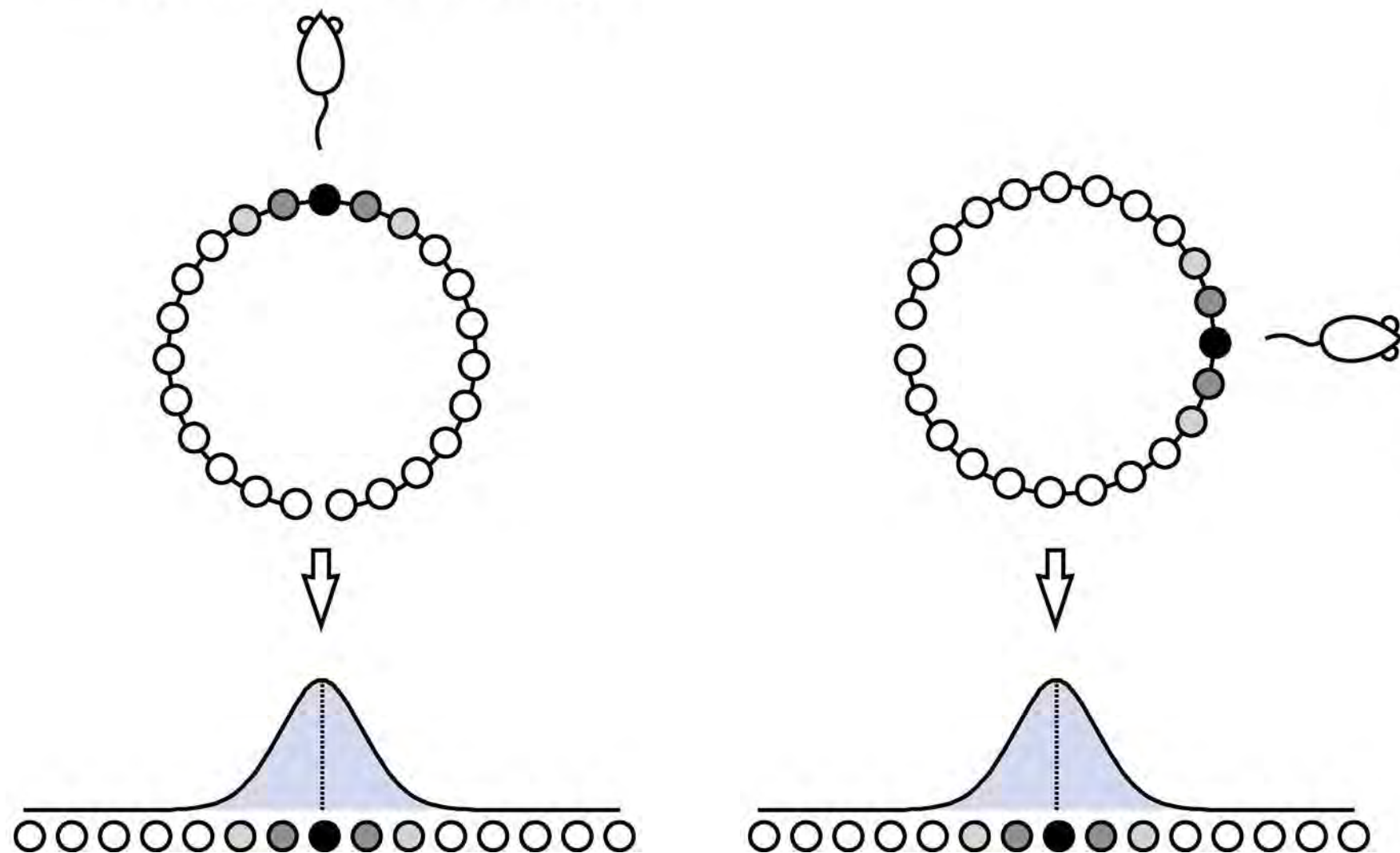
Tuning Curve of a Head-Direction Cell



Preferred Direction of a Single Head-Direction Cell



A ring attractor model

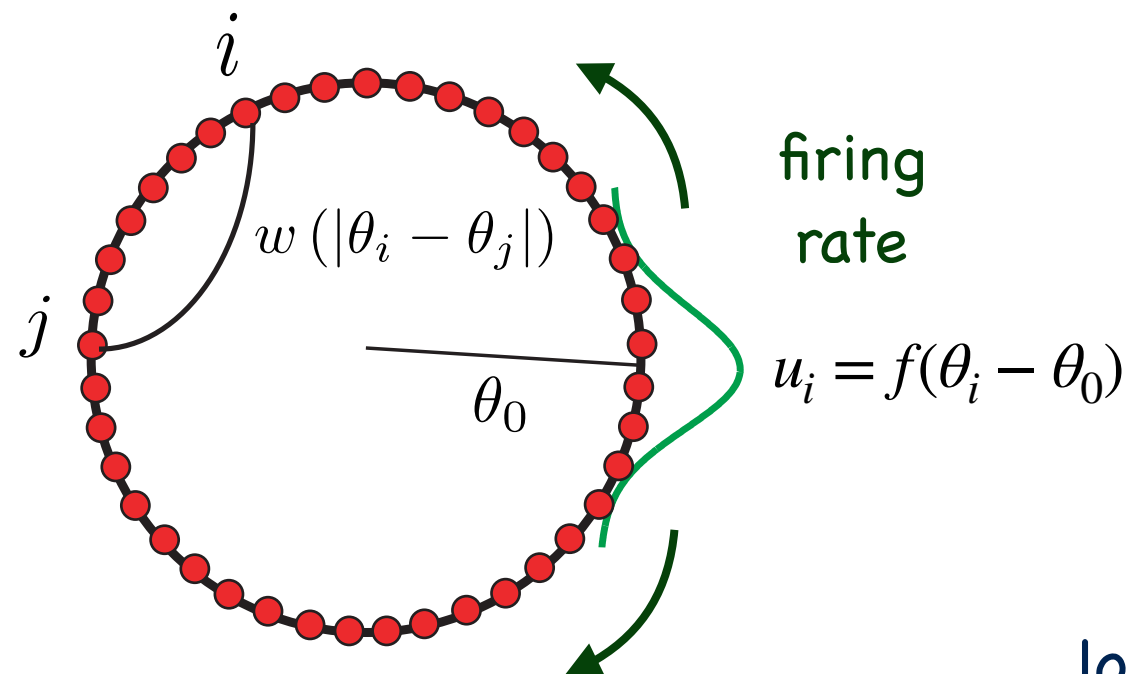


Ben-Yishai, Bar-Or, Sompolinsky, PNAS (1995)

Zhang, J. Neurosci. (1998)

Xie, Hahnloser, Seung, Phys. Rev. E (2002)

A ring attractor model



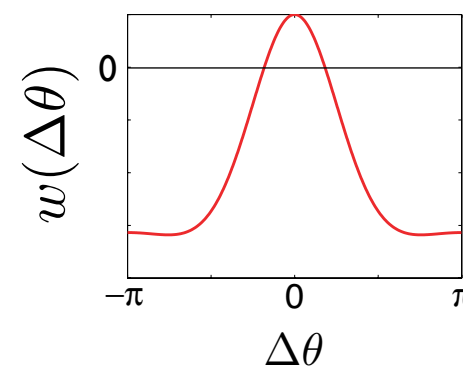
Solution:
A continuum of steady states
encode a continuous parameter (angle)

e.g., $u(\theta) = a + b \cos(\theta - \theta_0)$

local excitation, long range inhibition

$$\tau \frac{du_i}{dt} = -u_i + \left[\sum_{j=1..N} w_{ij} u_j + I_i^0 \right]_+$$

$$w_{ij} = w_{ji} = \frac{1}{N} J(|\theta_i - \theta_j|)$$

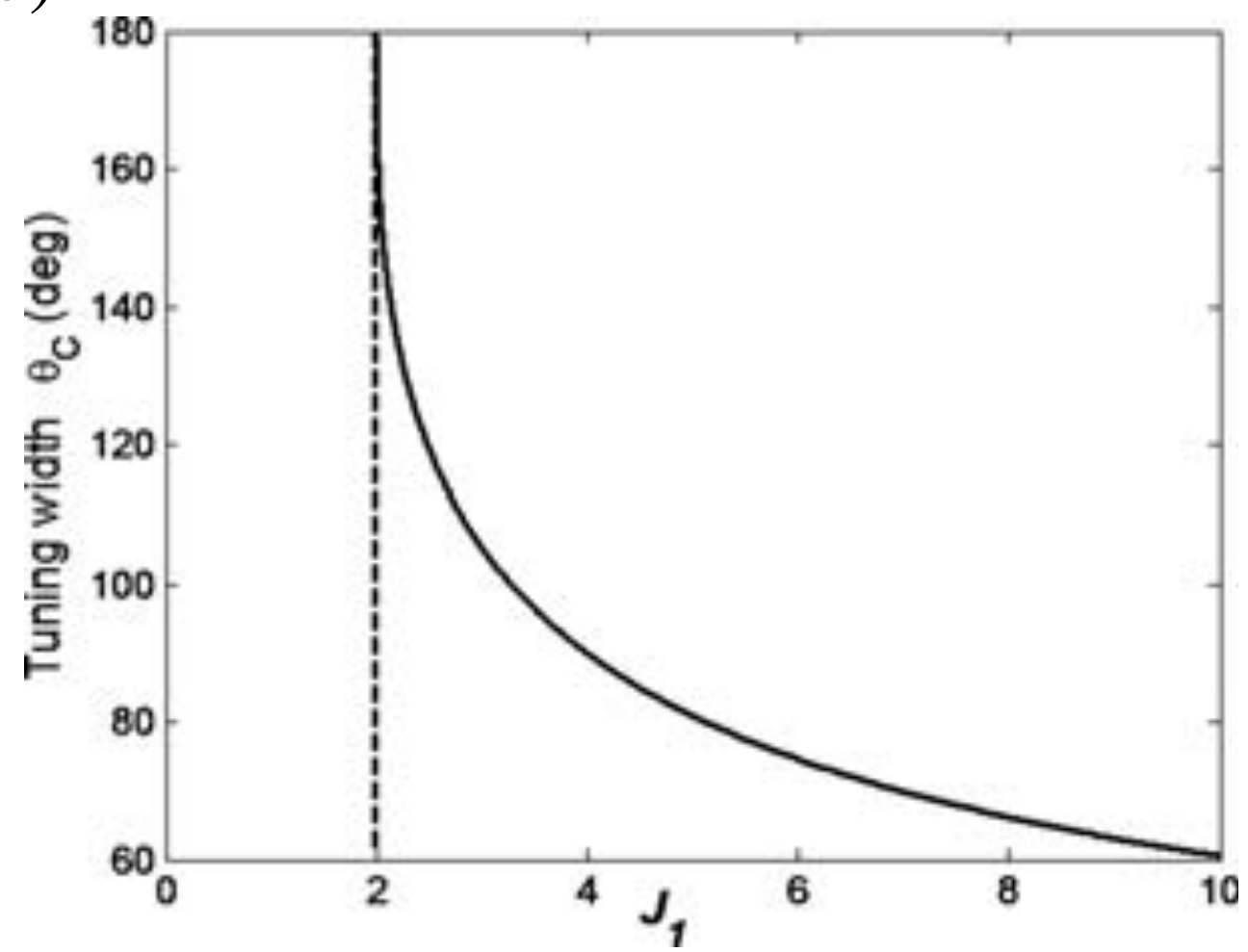


$$J = J_0 + J_1 \cos \theta$$

Let $N \rightarrow \infty$

$$\tau \frac{\partial u(\theta, t)}{\partial t} = -u(\theta, t) + \left[\frac{1}{2\pi} \int_{-\pi}^{\pi} J(\theta - \theta') u(\theta', t) d\theta' + I^0(\theta) \right]_+$$

\\



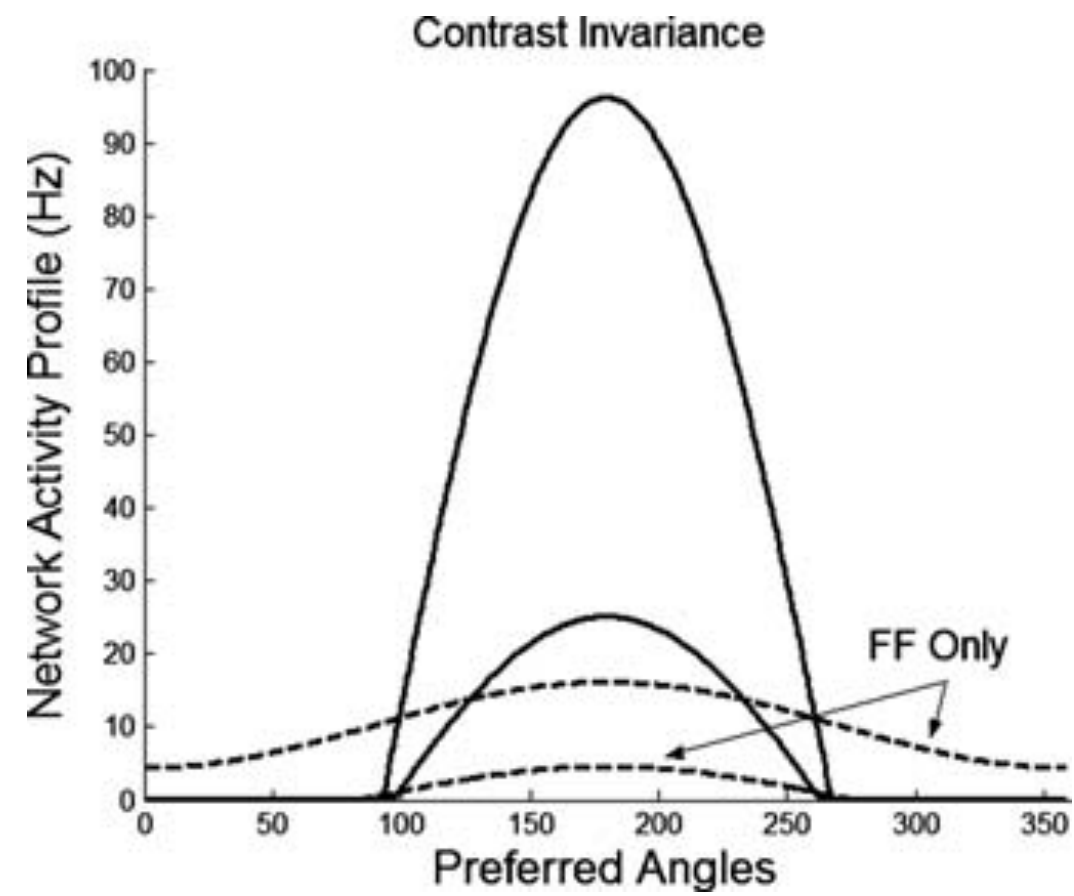
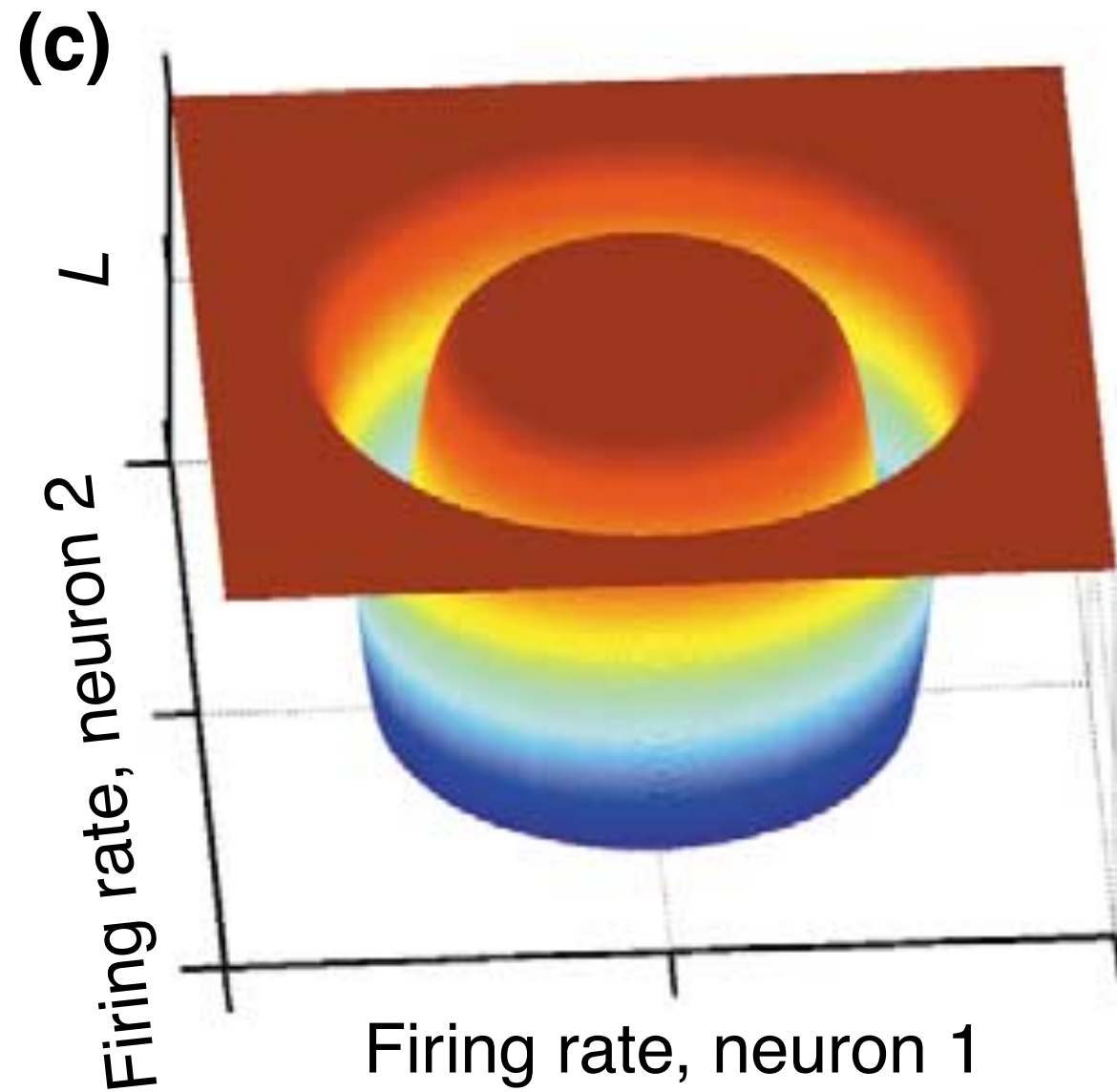


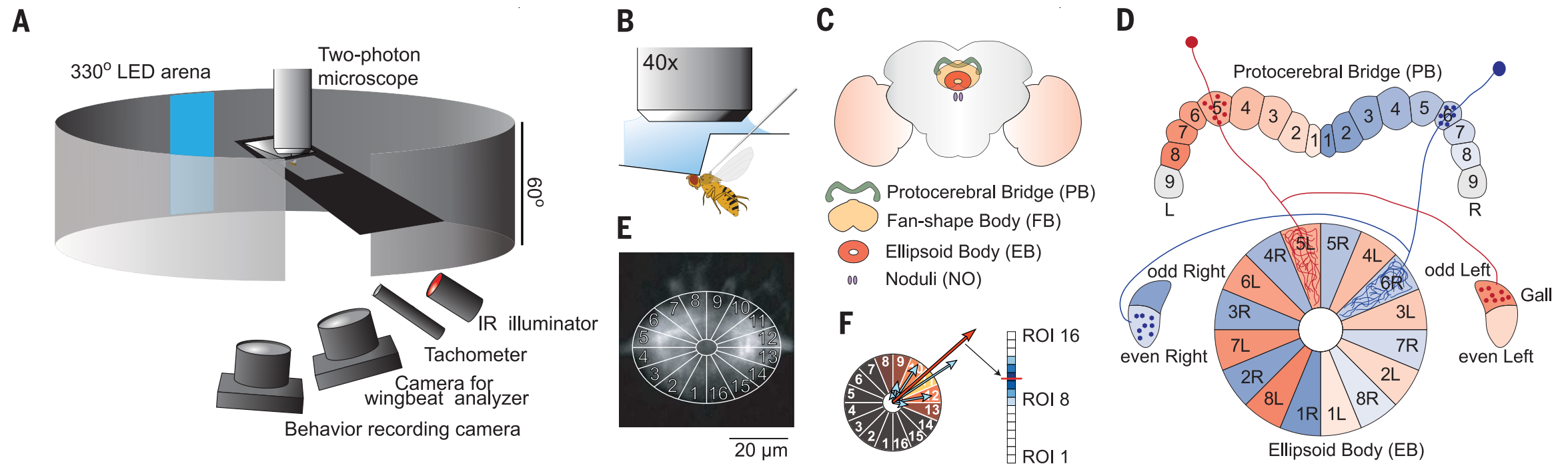
Fig. 5.2. Activity profile of the ring network for different contrasts.

Energy landscape of a continuous attractor



Kepecs and Brody, Current Opinion in Neurobiology 2003

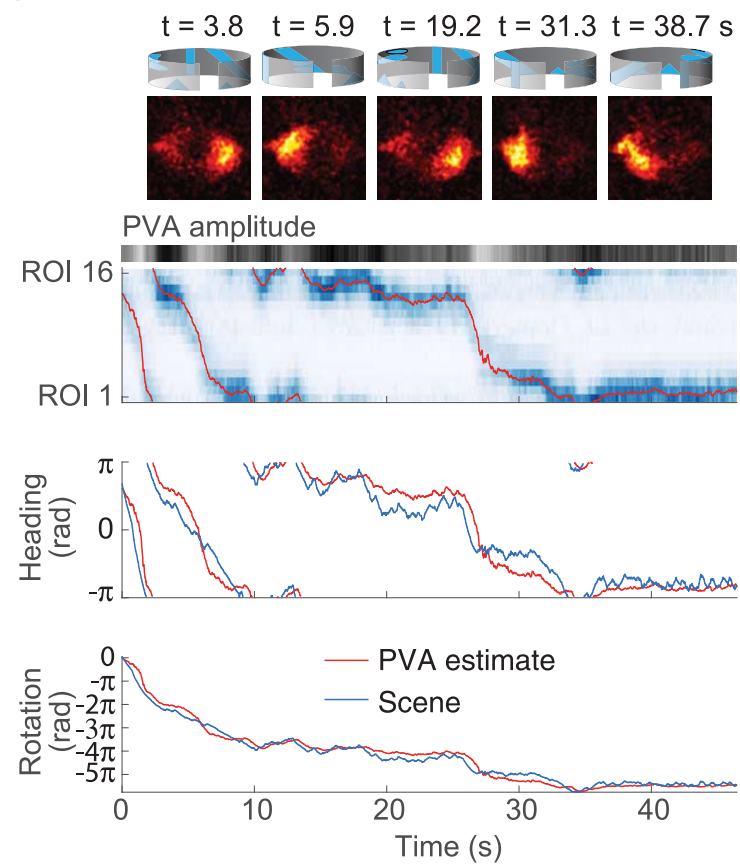
Ring attractor dynamics in fly brain



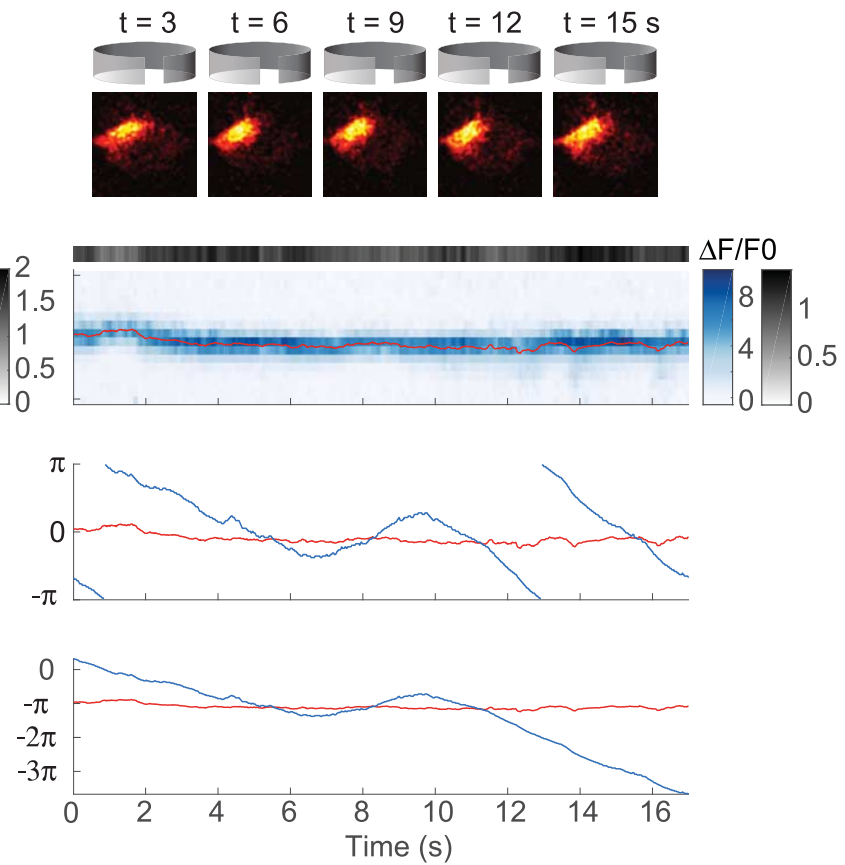
Kim *et al.*, *Science* **356**, 849–853 (2017)

Ring attractor dynamics in fly brain

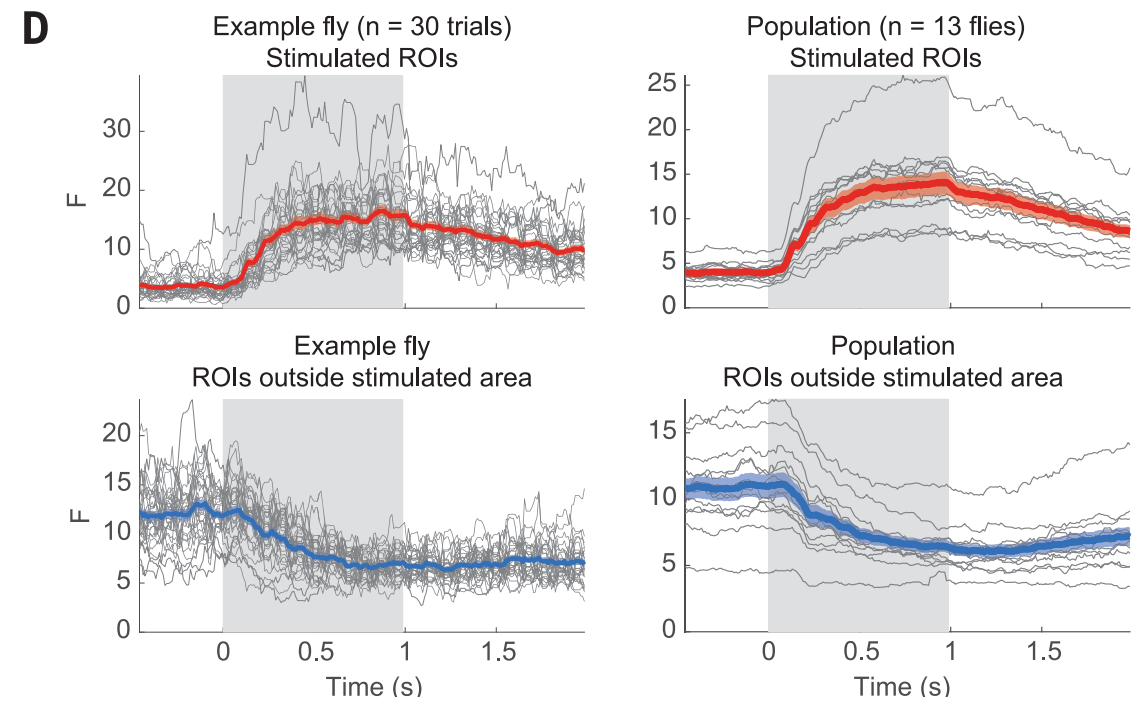
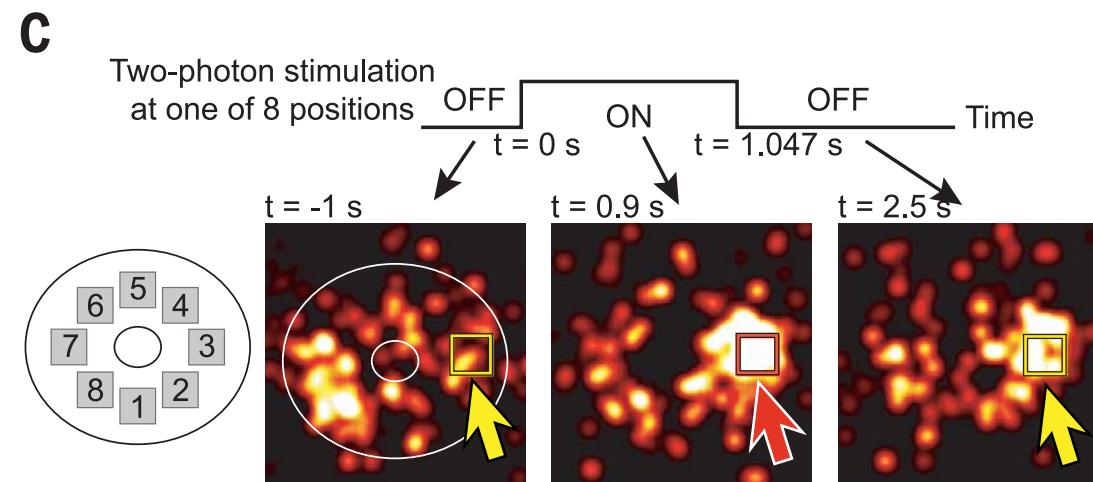
G



H



Ring attractor dynamics in fly brain



Memory recall and Hopfield network

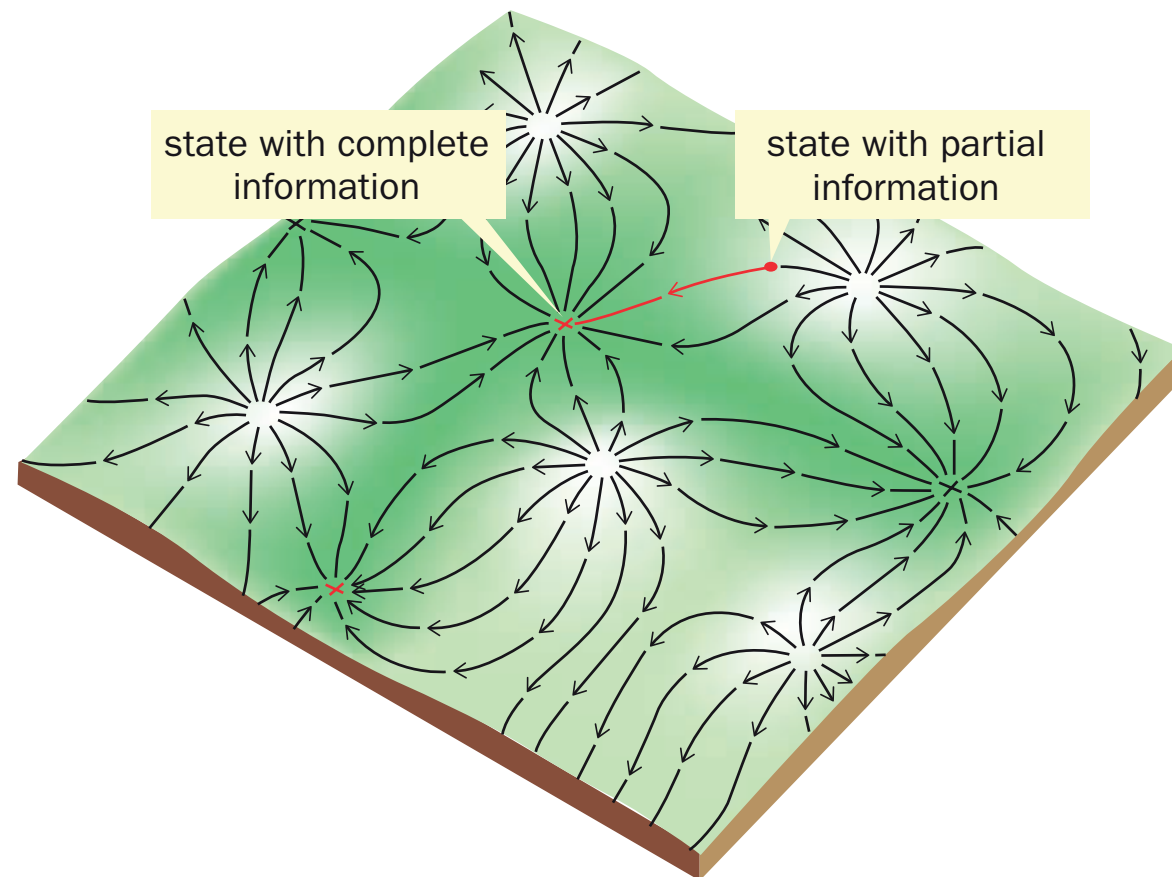


FIGURE 5.51

Recall of memories in the Hopfield model. The number of patterns K stored in the network is proportional to the number of neurons, so that $\alpha = K/N$ is the natural parameter. As α increases, the fraction of errors in the recalled patterns increases, until at some critical value α_c recall fails catastrophically, and the states of the network are not correlated with the stored patterns at all. Redrawn from Amit et al. (1985).

